

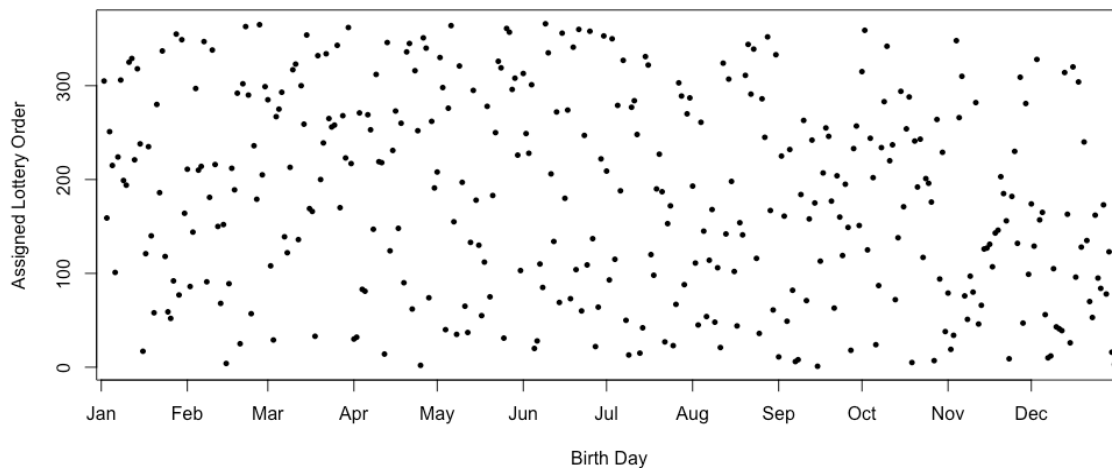
Math 258- Categorical Data Analysis
Problem Set 7
Analysis of the Vietnam War Draft Lottery

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In December 1969, a lottery system, based on birthdays, was used to determine the order of call to military service in the Vietnam war. In this system, the days of each month were written on a slip of paper and stored in a capsule. Capsules for each month were poured and pushed aside (in monthly order) in a large wooden box which was then shook several times. Lottery number was then assigned to the birth date in order in which the capsules were drawn.

We are curious whether this system is fair or if the setup has caused some association between the birth date and the lottery number assigned. First, we visualize the raw data (Figure 1), which shows the relationship between the draftee's day of birth and assigned lottery number.

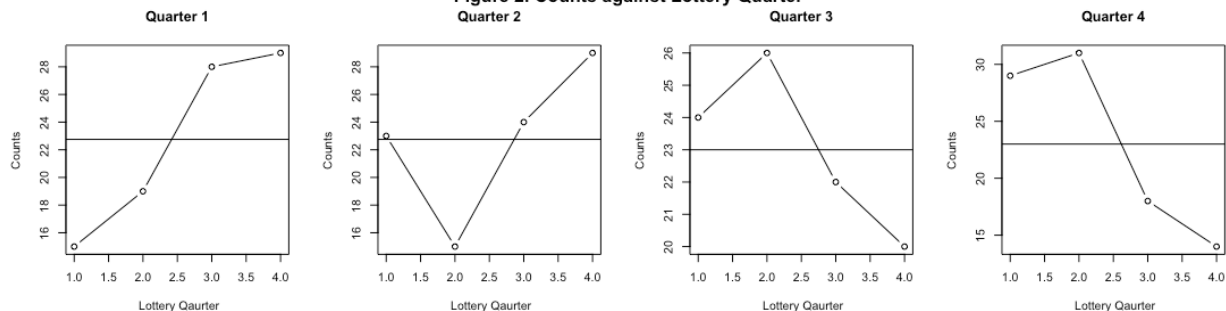
Figure 1. Birth Day against Lottery Order



As the points are scattered with no apparent pattern, it seems the lottery number is unrelated to the draftees' birthday. Perhaps the lottery process used was sufficiently random and fair.

To further analyze the data we combine the results into four quarters. The moment we visualize the four quarters (Figure 2), we observe evidence that the system is perhaps not fair after all. For instance, the first and the last plots clearly show that the counts are low for the first half and high for the second half for Quarter 1 and vice versa for Quarter 4.

Figure 2. Counts against Lottery Quarter



Hence, to assess the fairness of the lottery process, we start by conducting a Chi-Squared Test of Independence between birth quarter and lottery quarter. First, the assumptions should be checked prior to running the test.^[1] The assumption that the levels of the two categorical variables are mutually exclusive is satisfied because each person has a unique birthday and each birthday is assigned a unique lottery number. Another assumption is that the expected cell count should be 5 or more in at least 80% of the cells, and no cell should have an expected of less than 1. This assumption is most likely to be met if the sample size equals at least the number of cells multiplied by 5.^[2] Here there are 16 cells and the sample size of 366 is greater than $16 * 5 = 80$. Thus, this assumption is also verified.

The Chi-Square test yields a Pearson statistic of 20.44 with 9 degrees of freedom, which is greater than the critical value of 16.92. The null hypothesis of independence between birth quarter and lottery quarter is rejected at a 5% significance level. We have sufficient evidence to conclude that there is an association between the quarter of a year a birthday falls and the lottery quarter to which they were randomly assigned.

To get an overall idea of association, we use Goodman-Kruskal Gamma which in this case is -0.22 . This gives us an idea that perhaps, if we compare the lottery quarter of someone from the last two quarters to someone from the first two, they will most likely be assigned with small and large lottery numbers, respectively.

We want to see which cells caused the Chi-Square Test to produce a significant result, so we examine the standardized Pearson residuals. The table on the left highlights the residuals whose absolute value is greater than 2. These four cells contributed most to the rejection of H_0 in the Chi-Square Test. Three of these four cells are negative and reside on the diagonal, which implies that much of the apparent association between Birth Quarter and Lottery Quarter can be attributed to Birth Quarter groups being under-represented in their respective Lottery Quarters. The table on the right highlights the residuals which are negative, which continues to tell the same story as before. We see that whatever quarter you are born in, you're less likely than expected to be drafted in that quarter. We also see that, in general, people born in later quarters tend to be drafted more frequently than expected in earlier quarters, and people born in earlier quarters tend to be drafted more frequently than expected in later quarters.

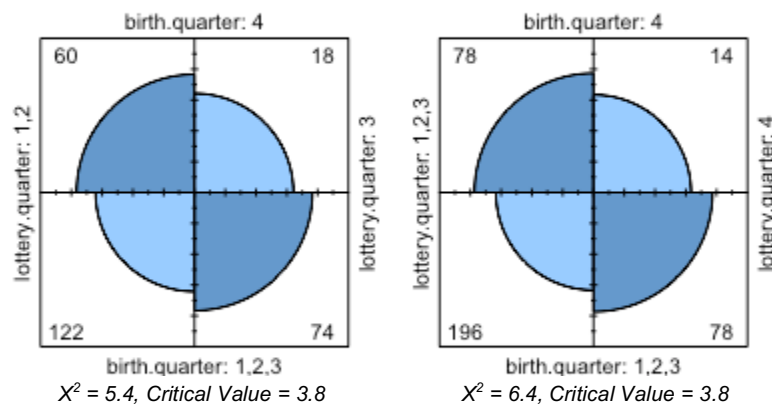
Figure 3. Standardized Pearson Residuals

Birth Quarter	Lottery Quarter			
	1	2	3	4
1	-2.13	-1.01	1.43	1.71
2	0.10	-2.13	0.31	1.71
3	0.31	0.87	-0.31	-0.87
4	1.71	2.27	-1.42	-2.53

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An exact partitioning of the 4x4 contingency table into nine 2x2 tables further reveals where the deviation is occurring. A Chi-Square test of independence fails to establish an association between Birth Quarter and Lottery Quarter at a 5% significance level in 7 out of 9 of the partitioned tables. The two tables which did produce a significant Chi-Square test statistic both split the birth groups into quarter 4 by itself and quarters 1, 2, and 3 combined. The following fourfold plots (Figure 4) illustrate the deviation that occurs in these two tables.

Figure 4. Fourfold plots.



This shows that those born in quarter 4 are more likely than those born in the other three quarters to be drafted in quarters 1 or 2 than in quarter 3, and are more likely to be drafted in quarters 1, 2, or 3 than in quarter 4. In other words, the randomization process of the draft lottery was most unfair to those born in October, November, and December because their draft order tended to be earlier than we would expect to see if the randomization was fair.

In order to further illustrate the difference between those born early in the year and those born late in the year, we condense the contingency table down to a 2x2 table. We find that among those born in the second half of the year, about 60% are drafted in the first half. Conversely, among those born in the first half of the year, about 60% are drafted in the second half. A 95% Wald Confidence Interval for the difference of proportions is [0.1019, 0.3026]. If we had a group of 100 people born in the second half of the year and a group of 100 people born in the first half of the year, we expect about 10 to 30 more people in the first group to be selected in the first half of the draft than those in the second group.

At the beginning of the analysis, just by observing the raw data, it seemed that the lottery system was randomly and fairly calling people for service in the war. But, as soon as we visualized the quarterly results, it was clear that the system was perhaps not fair. This intuition is then confirmed by our analysis. After concluding dependency between the Birth Quarter and Lottery Quarter, we further explored that for a person is least likely to be called for service in the quarter in which he was born. We also see that people born between January and March are mostly called for service in the last three months of the year and those who are born between October and December are called in the first three months. Similarly, we see that people born in the first half are more likely to be called in the second half for service. This would imply, for people born in the same year, the lottery system favors younger people, as it calls them to service first. Hence, we conclude that the lottery system is unfair.

References

- [1]Mchugh M, The Chi-square test of independence, 2013;
<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3900058>
- [2]Bewick V, Cheek L, Ball J. Statistics review 8: Qualitative data - tests of association. Crit Care. 2004;8:46–53. <http://dx.doi.org/10.1186/cc2428>.

Appendix

```
lottery<-read.table("draftLottery1970.txt",header=T)
#-----
#scatter plot
#-----
days=c(0,31,29,31,30,31,30,31,31,30,31,30,31)
plot(lottery[,1],lottery[,2],xaxt="n",xlim=c(12,354),pch=16,cex=.7,xlab="Birth Day",ylab="Assigned Lottery Order",main="Figure 1. Birth Day against Lottery Order")
axis(1,at=cumsum(days)[-13],labels =
c("Jan","Feb","Mar","Apr","May","Jun","Jul","Aug","Sep","Oct","Nov","Dec"))

#-----
# 1st quarter
#-----
par(mfrow = c(1,4))
q1<-c(15,19, 28, 29)
plot(1:4, q1, xlab=" Lottery Qaurter", ylab="Counts", main= "Quarter 1", type = "b")
abline(91/4,0)
#-----
# 2st quarter
#-----
q2<-c(23, 15, 24, 29)
plot(1:4, q2, xlab=" Lottery Qaurter", ylab="Counts", main= "Quarter 2", type = "b")
abline(91/4,0)
#-----
# 3st quarter
#-----
q3<-c(24, 26, 22, 20)
plot(1:4, q3, xlab=" Lottery Qaurter", ylab="Counts", main= "Quarter 3", type = "b")
abline(92/4,0)
#-----
# 4st quarter
#-----
q4<-c(29 ,31 ,18, 14)
plot(1:4,q4,xlab="Lottery Qaurter",ylab="Counts",main="Quarter 4",type="b")
abline(92/4,0)
```

```

#-----
#chi-square test
#-----
daysInQtr <- c(91, 91, 92, 92)
cutpts <- c(1, cumsum(daysInQtr))
lottery.qtr <- cut(lottery$draft.order, cutpts, include.lowest=T, labels = 1:4)
birth.qtr <- rep(1:4, daysInQtr)
lottery.table.2 <- table(birth.quarter=birth.qtr, lottery.quarter =
lottery.qtr)
print(lottery.table.2)
cat("\nRow totals:", rowSums(lottery.table.2), "\n")
cat("Col totals:", colSums(lottery.table.2), "\n\n")
chisq.indep(lottery.table.2)

#-----
#goodman-kruskal gamma
#-----
gk.gamma(lottery.table.2)

#-----
#residuals
#-----
round(chisq.indep(lottery.table.2)$std.res,2)

#-----
#exact partitioning
#-----
partition.1=lottery.table.2[1:2,1:2]
partition.2=as.table(matrix(c(sum(lottery.table.2[1,1:2]),sum(lottery.table.2
[2,1:2]),sum(lottery.table.2[1,3]),sum(lottery.table.2[2,3])),nrow=2))
dimnames(partition.2)=list(birth.quarter=c("1", "2"), lottery.quarter=c("1,2", "
3"))
partition.3=as.table(matrix(c(sum(lottery.table.2[1,1:3]),sum(lottery.table.2
[2,1:3]),sum(lottery.table.2[1,4]),sum(lottery.table.2[2,4])),nrow=2))
dimnames(partition.3)=list(birth.quarter=c("1", "2"), lottery.quarter=c("1,2,3",
"4"))
partition.4=as.table(matrix(c(sum(lottery.table.2[1:2,1]),sum(lottery.table.2
[3,1]),sum(lottery.table.2[1:2,2]),sum(lottery.table.2[3,2])),nrow=2))
dimnames(partition.4)=list(birth.quarter=c("1,2", "3"), lottery.quarter=c("1", "
2"))
partition.5=as.table(matrix(c(sum(lottery.table.2[1:2,1:2]),sum(lottery.table
.2[3,1:2]),sum(lottery.table.2[1:2,3]),sum(lottery.table.2[3,3])),nrow=2))

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```

dimnames(partition.5)=list(birth.quarter=c("1,2", "3"), lottery.quarter=c("1,2",
"3"))
partition.6=as.table(matrix(c(sum(lottery.table.2[1:2,1:3]),sum(lottery.table
.2[3,1:3]),sum(lottery.table.2[1:2,4]),sum(lottery.table.2[3,4])),nrow=2))
dimnames(partition.6)=list(birth.quarter=c("1,2", "3"), lottery.quarter=c("1,2,
3", "4"))
partition.7=as.table(matrix(c(sum(lottery.table.2[1:3,1]),sum(lottery.table.2
[4,1]),sum(lottery.table.2[1:3,2]),sum(lottery.table.2[4,2])),nrow=2))
dimnames(partition.7)=list(birth.quarter=c("1,2,3", "4"), lottery.quarter=c("1"
,"2"))
partition.8=as.table(matrix(c(sum(lottery.table.2[1:3,1:2]),sum(lottery.table
.2[4,1:2]),sum(lottery.table.2[1:3,3]),sum(lottery.table.2[4,3])),nrow=2))
dimnames(partition.8)=list(birth.quarter=c("1,2,3", "4"), lottery.quarter=c("1,
2", "3"))
partition.9=as.table(matrix(c(sum(lottery.table.2[1:3,1:3]),sum(lottery.table
.2[4,1:3]),sum(lottery.table.2[1:3,4]),sum(lottery.table.2[4,4])),nrow=2))
dimnames(partition.9)=list(birth.quarter=c("1,2,3", "4"), lottery.quarter=c("1,
2,3", "4"))
#checking to make sure we have a true partition
chisq.indep(partition.1)$lr+chisq.indep(partition.2)$lr+chisq.indep(partition
.3)$lr+chisq.indep(partition.4)$lr+chisq.indep(partition.5)$lr+chisq.indep(pa
rtition.6)$lr+chisq.indep(partition.7)$lr+chisq.indep(partition.8)$lr+chisq.i
ndep(partition.9)$lr;chisq.indep(lottery.table.2)$lr #both are 21.10638
chisq.indep(partition.1) #fail to reject
chisq.indep(partition.2) #fail to reject
chisq.indep(partition.3) #fail to reject
chisq.indep(partition.4) #fail to reject
chisq.indep(partition.5) #fail to reject
chisq.indep(partition.6) #fail to reject (close though)
chisq.indep(partition.7) #fail to reject
chisq.indep(partition.8) #reject
chisq.indep(partition.9) #reject

#-----
#fourfold plots
#-----
par(mfrow=c(1,2))
partition.8=partition.8[2:1,1:2]
fourfoldplot(partition.8,conf.level=0)
partition.9=partition.9[2:1,1:2]
fourfoldplot(partition.9,conf.level=0)

```

```

#-----
#difference of proportions
#-----
daysInHalf <- c(182,184)
cutpts <- c(1, cumsum(daysInHalf))
lottery.half <- cut(lottery$draft.order, cutpts, include.lowest = TRUE,
labels = 1:2)
birth.half <- rep(1:2, daysInHalf)
lottery.table.3 <- table(birth.half=birth.half, lottery.half = lottery.half)
print(lottery.table.3)
lottery.table.3=lottery.table.3[2:1,1:2]
lottery.table.3/sum(lottery.table.3)
t1hat=lottery.table.3[1,1]/sum(lottery.table.3[1,])
t2hat=lottery.table.3[2,1]/sum(lottery.table.3[2,])
DPhat=t1hat-t2hat
var=t1hat*(1-t1hat)/sum(lottery.table.3[1,])+t2hat*(1-
t2hat)/sum(lottery.table.3[2,])
DPhat-qnorm(.975)*sqrt(var)
DPhat+qnorm(.975)*sqrt(var)

```